Class Activity: 10.07.2026

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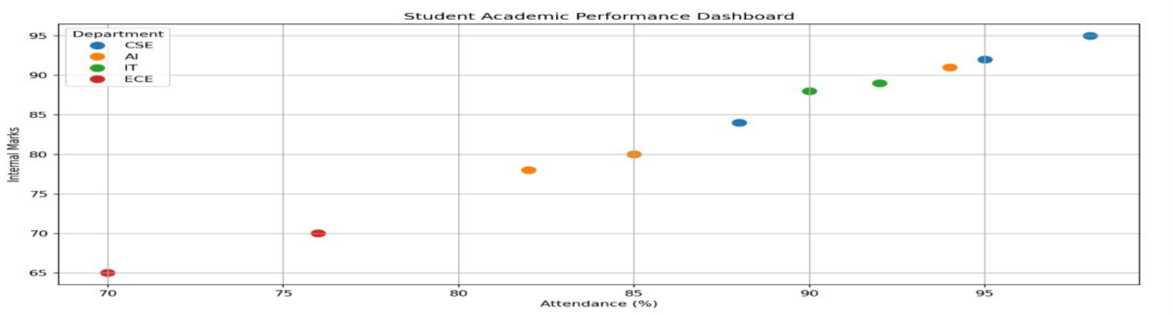
Scenario 1: Student Academic Performance Dashboard

Code:

```
# Create Dataset
student_data <- data.frame(
  Student_ID = c("S101","S102","S103","S104","S105",
               "S106","S107","S108","S109","S110"),
  Attendance = c(95,82,90,76,98,85,92,70,88,94),
  Internal_Marks = c(92,78,88,70,95,80,89,65,84,91),
  Department = c("CSE","AI","IT","ECE","CSE",
               "AI","IT","ECE","CSE","AI")
)
print(student_data)
colors <- c("CSE"="red","AI"="blue","IT"="green","ECE"="purple")
plot(student_data$Attendance,
     student_data$Internal_Marks,
     col=colors[student_data$Department],
     pch=19,
     cex=1.5,
     xlab="Attendance (%)",
     ylab="Internal Marks",
     main="Student Academic Performance Dashboard")
legend("topleft",
       legend=names(colors),
       col=colors,
       pch=19,
       title="Department")
```

Output:



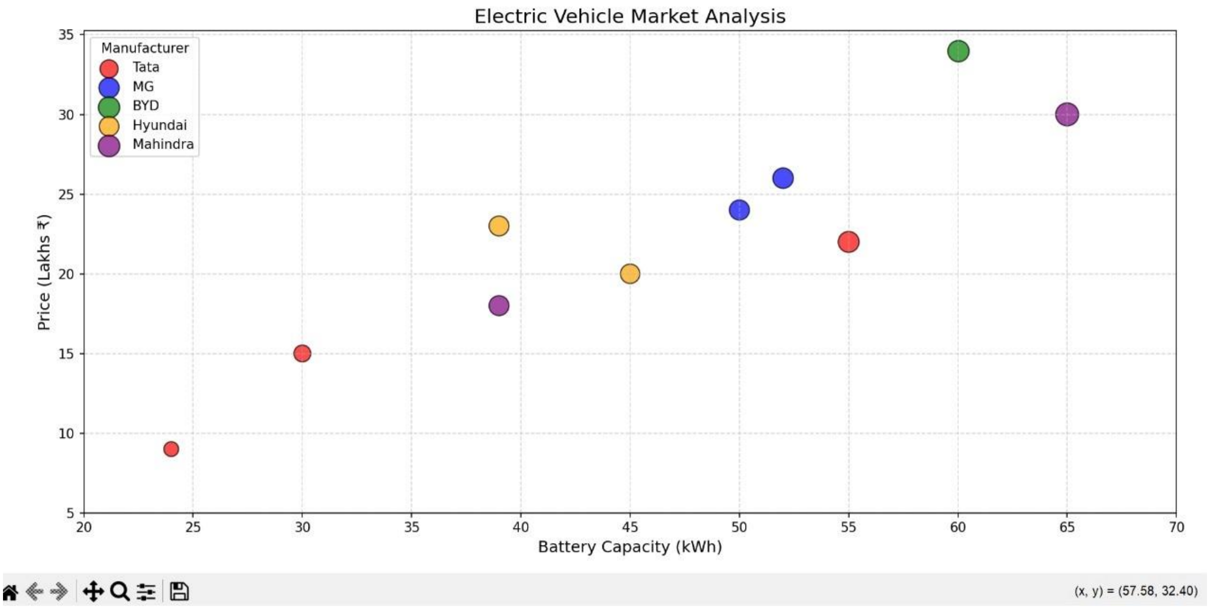
Scenario 2: Electric Vehicle Market Analysis

Code:

```
vehicle <- data.frame(
  Model=c("Nexon EV","Tiago EV","MG ZS EV","BYD Atto 3",
          "Kona Electric","XUV400","Curvv EV",
          "BE6","Creta EV","Windsor EV"),
  Battery=c(30,24,50,60,39,39,55,65,45,52),
  Price=c(15,9,24,34,23,18,22,30,20,26),
  Range=c(325,250,461,521,452,456,500,600,430,480),
  Manufacturer=c("Tata","Tata","MG","BYD","Hyundai",
                 "Mahindra","Tata","Mahindra",
                 "Hyundai","MG")
)
cols <- c("Tata"="red",
          "MG"="blue",
          "BYD"="green",
          "Hyundai"="orange",
          "Mahindra"="purple")
plot(vehicle$Battery,
     vehicle$Price,
     pch=19,
     cex=vehicle$Range/200,
     col=cols[vehicle$Manufacturer],
     xlab="Battery Capacity (kWh)",
```

```
    ylab="Price (Lakhs ₹)",
    main="Electric Vehicle Market Analysis")
legend("topleft",
       legend=names(cols),
       col=cols,
       pch=19)
```

Output:



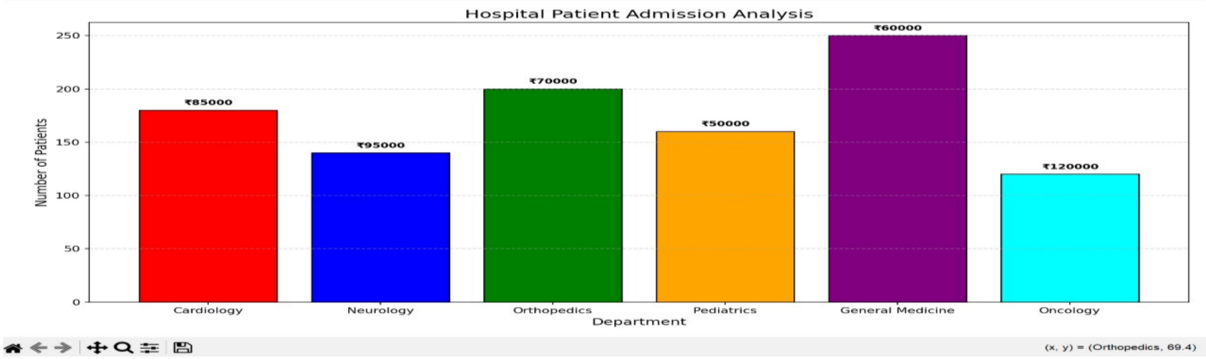
Scenario 3: Hospital Patient Admission Analysis

Code:

```
hospital <- data.frame(
  Department=c("Cardiology","Neurology","Orthopedics",
               "Pediatrics","General Medicine","Oncology"),
  Patients=c(180,140,200,160,250,120),
  Cost=c(85000,95000,70000,50000,60000,120000)
)
bar <- barplot(hospital$Patients,
               names.arg=hospital$Department,
               col=c("red","blue","green",
                     "orange","purple","cyan"),
               ylim=c(0,280),
```

```
        main="Hospital Patient Admission Analysis",
        ylab="Number of Patients")
text(bar,
    hospital$Patients+10,
    labels=paste("₹",hospital$Cost,sep=""),
    cex=0.8)
```

Output:



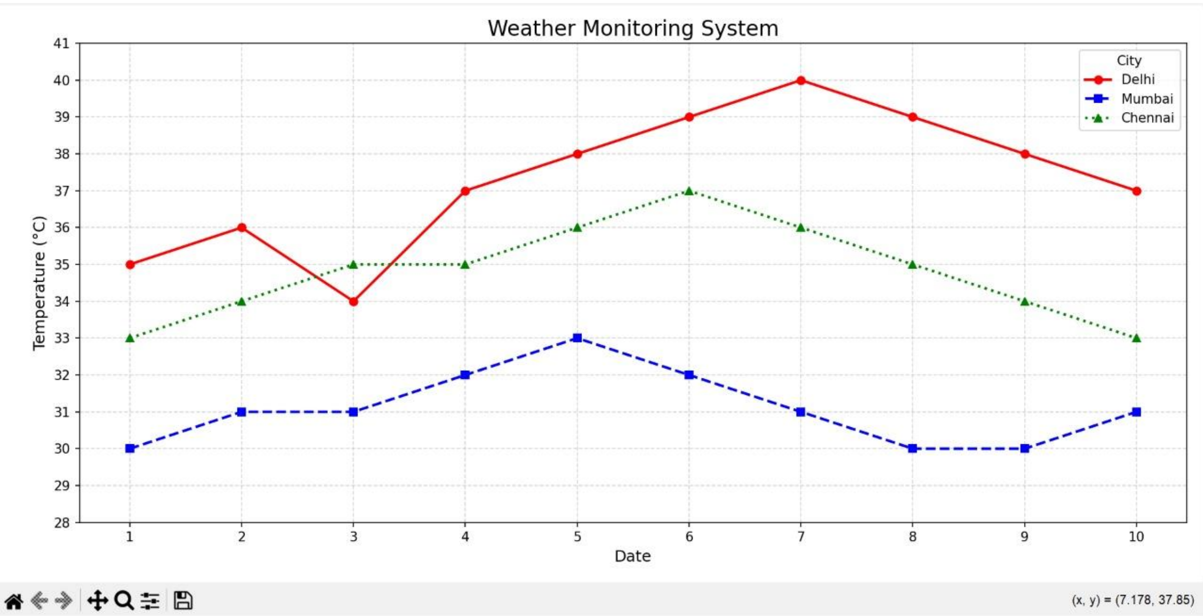
Scenario 4: Weather Monitoring System

Code:

```
date <- 1:10
delhi <- c(35,36,34,37,38,39,40,39,38,37)
mumbai <- c(30,31,31,32,33,32,31,30,30,31)
chennai <- c(33,34,35,35,36,37,36,35,34,33)
plot(date,delhi,
    type="o",
    col="red",
    lty=1,
    ylim=c(28,42),
    xlab="Date",
    ylab="Temperature (°C)",
    main="Weather Monitoring System")
lines(date,mumbai,
     type="o",
```

```
      col="blue",
      lty=2)
lines(date,chennai,
      type="o",
      col="green",
      lty=3)
legend("topleft",
       legend=c("Delhi","Mumbai","Chennai"),
       col=c("red","blue","green"),
       lty=c(1,2,3),
       pch=1)
```

Output:



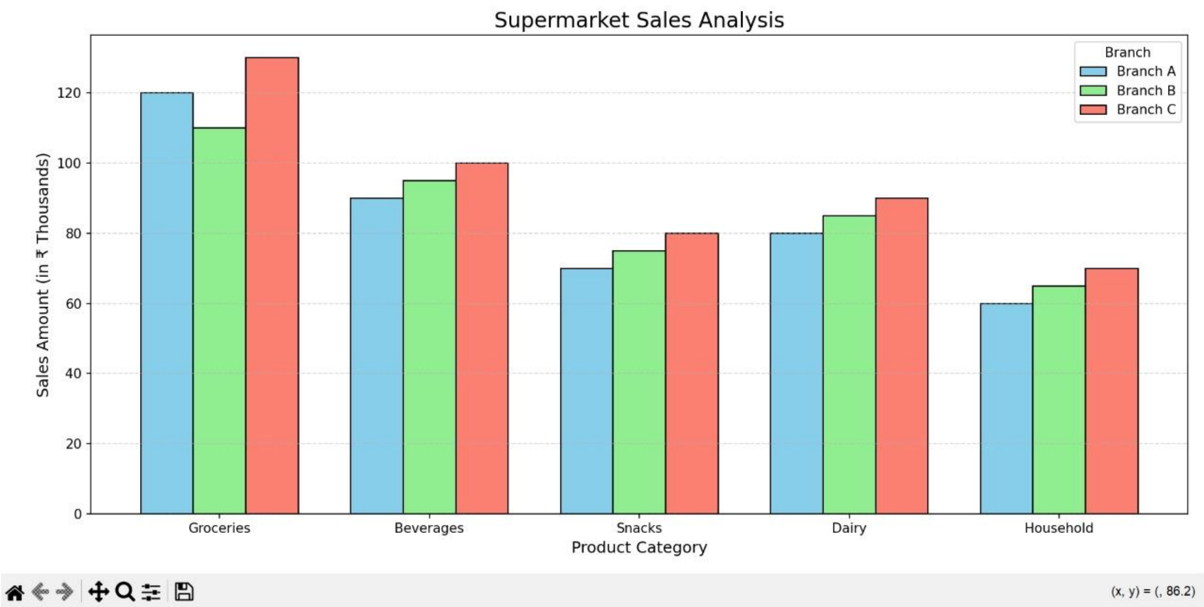
Scenario 5: Supermarket Sales Analysis

Code:

```
sales <- matrix(c(
120,110,130,
90,95,100,
70,75,80,
80,85,90,
60,65,70),
```

```
nrow=3,
byrow=TRUE)
colnames(sales) <- c("Groceries","Beverages","Snacks")
rownames(sales) <- c("Branch A","Branch B","Branch C")
barplot(sales,
    beside=TRUE,
    col=c("skyblue","lightgreen","salmon"),
    legend=rownames(sales),
    xlab="Product Category",
    ylab="Sales Amount",
    main="Supermarket Sales Analysis") ()
```

Output:



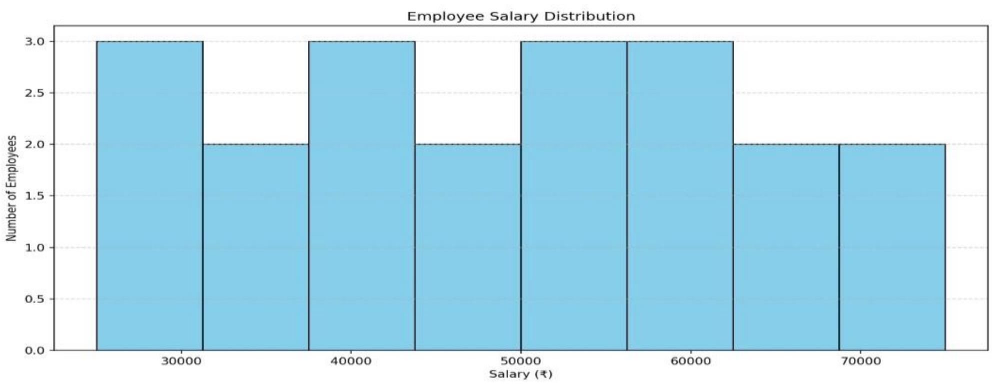
Scenario 6: Employee Salary Distribution

Code:

```
salary<-c(
25000,28000,30000,32000,35000,
38000,40000,42000,45000,47000,
50000,52000,55000,58000,60000,
62000,65000,68000,70000,75000
)
```

```
hist(
salary,
breaks=8,
col="skyblue",
border="black",
main="Employee Salary Distribution",
xlab="Salary (₹)",
ylab="Number of Employees"
)
grid()
```

Output:



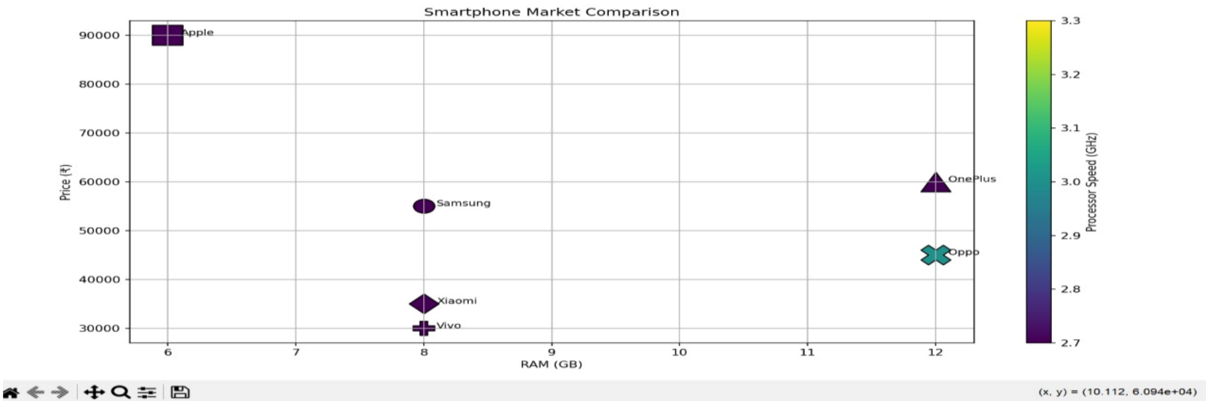
Scenario 7: Smartphone Market Comparison

Code:

```
phone<-data.frame(
Brand=c("Samsung","Apple","OnePlus","Xiaomi","Vivo","Oppo"),
RAM=c(8,6,12,8,8,12),
Processor=c(2.8,3.2,3.1,2.7,2.6,3.0),
Price=c(55000,90000,60000,35000,30000,45000),
Storage=c(128,256,256,128,128,256)
)
colors<-c("red","blue","green","orange","purple","brown")
symbols<-c(16,17,15,18,8,3)
```

```
plot(
phone$RAM,
phone$Price,
type="n",
xlab="RAM (GB)",
ylab="Price (₹)",
main="Smartphone Market Comparison"
)
for(i in 1:nrow(phone)){
points(
phone$RAM[i],
phone$Price[i],
pch=symbols[i],
col=colors[i],
cex=phone$Storage[i]/100
)
text(
phone$RAM[i]+0.2,
phone$Price[i],
labels=phone$Brand[i],
cex=0.8
)
} ()
```

Output:



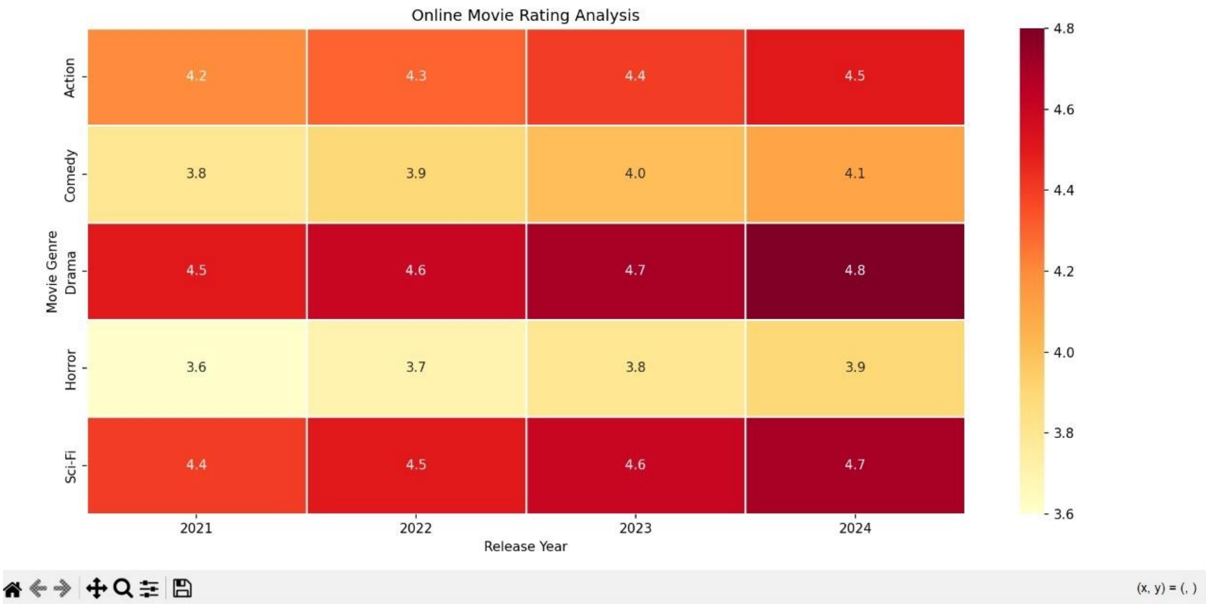
Scenario 8: Online Movie Rating Analysis

Code:

```
ratings<-matrix(
c(
4.2,4.3,4.4,4.5,
3.8,3.9,4.0,4.1,
4.5,4.6,4.7,4.8,
3.6,3.7,3.8,3.9,
4.4,4.5,4.6,4.7
),
nrow=5,
byrow=TRUE
)
rownames(ratings)<-c(
"Action",
"Comedy",
"Drama",
"Horror",
"Sci-Fi"
)
colnames(ratings)<-c(
"2021",
"2022",
"2023",
"2024"
)
heatmap(
ratings,
Rowv=NA,
Colv=NA,
```

```
col=heat.colors(20),
scale="none",
main="Online Movie Rating Analysis"
)
```

Output:



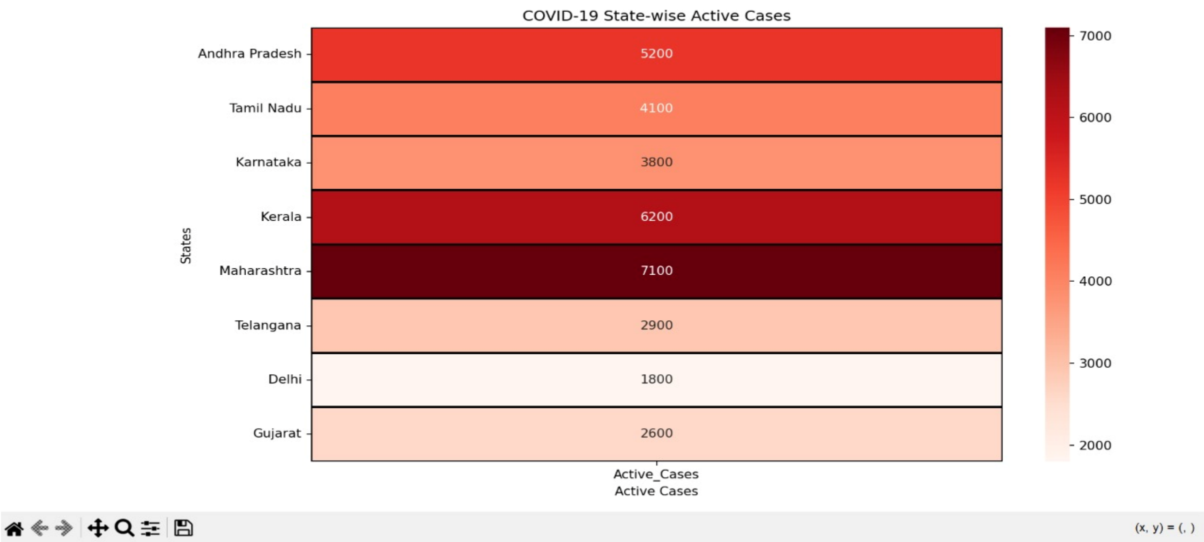
Scenario 9: COVID-19 State-wise Analysis

Code:

```
covid<-matrix(
c(
5200,
4100,
3800,
6200,
7100,
2900,
1800,
2600
),
nrow=8
```

```
)
rownames(covid)<-c(
"Andhra Pradesh",
"Tamil Nadu",
"Karnataka",
"Kerala",
"Maharashtra",
"Telangana",
"Delhi",
"Gujarat"
)
colnames(covid)<-"Active Cases"
heatmap(
covid,
Rowv=NA,
Colv=NA,
col=heat.colors(20),
scale="none",
main="COVID-19 State-wise Analysis"
) ()
```

Output:



Scenario 10: Iris Flower Classification

Code:

```
data(iris)
colors<-c(
"red",
"blue",
"green"
)
symbols<-c(
16,
17,
15
)
plot(
iris$Sepal.Length,
iris$Petal.Length,
type="n",
xlab="Sepal Length (cm)",
ylab="Petal Length (cm)",
main="Iris Flower Classification"
)
for(i in 1:3){
temp<-iris[iris$Species==levels(iris$Species)[i],]
points(
temp$Sepal.Length,
temp$Petal.Length,
col=colors[i],
pch=symbols[i],
cex=temp$Sepal.Width/2
)
```

```
}
legend(
"topleft",
legend=levels(iris$Species),
col=colors,
pch=symbols,
title="Species"
) ()
```

Output: